Model	Task	Levy/Holt (Directional)		
		Precision	Recall	**Δ-*Recall***
LLaMA	I	67.0	**68.4**	*0*
	I^{GenArg}	69.0	66.9	*-1.5*
	$I^{RandArg\downarrow}$	64.0	63.8	*-4.6*
	$I^{RandArg\uparrow}$	67.2	53.7	*-14.7*
GPT-3.5	I	62.4	**92.3**	*0*
	I^{GenArg}	65.1	75.7	*-16.6*
	$I^{RandArg\downarrow}$	65.5	66.5	*-25.8*
	$I^{RandArg\uparrow}$	68.8	55.3	*-37.0*
PaLM	I	72.8	**76.2**	*0*
	I^{GenArg}	79.8	50.8	*-25.4*
	$I^{RandArg\downarrow}$	69.5	58.7	*-17.5*
	$I^{RandArg\uparrow}$	70.8	52.4	*-23.8*

Table 4: Exp-2. Scoring model outputs in different argument-replacement tasks. We indicate the **highest** and lowest recall score across replacement settings. Recall decreases sharply across settings in all models.

private or otherwise infeasibly large to read manually, containing many facts and human opinions unknowable to both the user and modeler.

6 Experiment 2: Entities are Indices to Memory

In §5, we have established that propositional memory explains a significant portion of false positives in LLM inference predictions. In this section, we continue by showing the importance of named entities in the process of LLMs' memory recall.

As described in §3.3, we manipulate the entities with the I^{GenArg} generic argument replacement, and two random entity replacements, one with infrequent-entities $I^{RandArg\downarrow}$ and one with frequent-entities $I^{RandArg\uparrow}$ (examples in Table 3).

By replacing arguments constrained by type, entailment labels are maintained; however, new samples should contain novel strings not attested in pretrain corpora. We expect that an ideal, generalizing model would maintain its predictions across all conditions; a flawed model utilizing the *attestation bias* would predict fewer `Entail`, since entities no longer identify these statements in training.

6.1 Results

We report results across conditions in Table 4. We observe two phenomena across all three models, aligning with the above conjecture of "flaws."

First, we observe that all models' behavior significantly changes in the same way when original entities are replaced by either entity types or random real entities. Despite similar (or marginally increasing) precision across conditions, recall degrades sharply from original entities (I) (GPT-3.5 @92.3) to random frequent entities ($I^{RandArg\uparrow}$) (GPT-3.5 @55.3). Generic-argument I^{GenArg} performance also degrades in this way, showing that this is not a matter of poorly selected real entities, but rather a loss of information from the original dataset which models were using to answer questions.

Second, across the 3 models, we observe a significant difference in recall between the two real entity conditions $I^{RandArg\downarrow}$ and $I^{RandArg\uparrow}$, which are both composed of unattested statements, but involve entities that differ in typical corpus frequency. Infrequent entities ($I^{RandArg\downarrow}$) yield better generalization and a higher recall (GPT-3.5 @66.5) than frequent entities ($I^{RandArg\uparrow}$) (GPT-3.5 @55.3).

These findings corroborate those from §5, that LLMs use memory as part of language inference, and additionally show that these memories are recalled using named entities acting as indices. These experiments demonstrate that too much prior exposure to an entity may impede model generalization when that entity is discussed in novel inferences: the more a model has read about an entity during pretraining, the less capable it is of drawing novel natural language inferences involving it. This is the case even though the inferences do not require detailed knowledge of the entity.

Like §5, the effect is consistent across models, indicating LLM pretraining is responsible.

We show similar results on RTE-1 in Appendix B. Further, instructing LLMs to ignore propositional memory in Appendix C shows little change.

7 Experiment 3: Relative Frequency Bias

We continue the conditioning experiments from §5, now exploring the relative frequency bias. Sample labels for this bias are denoted by the model-agnostic Φ as described in §3.1. Φ labels the conformance of sample predicates to the bias: $\Phi_<$ means P is less corpus-frequent than H by a margin (positive class), $\Phi_>$ means P more frequent than H by the margin (negative class). To control for differences between datasets, the margin is set so that 1/3 of samples are classed as "roughly equal" ($\Phi_\approx$), which we discard.

Following the observations in §6, we further apply a generic-argument transformation to control for attestation, yielding $I^{GenArg}_{RandPrem}$. With the entities masked, models cannot recall propositional memory for this task: by re-calculating the Λ mea-

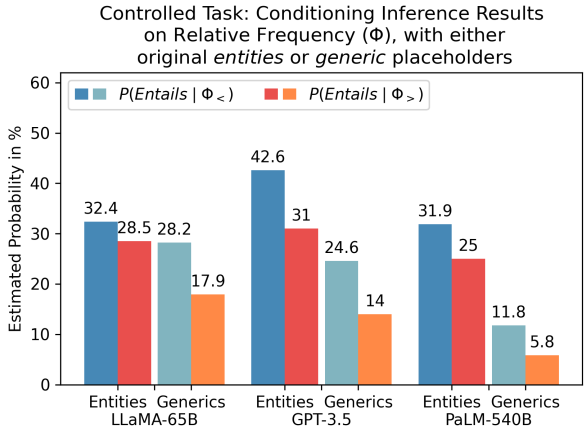

Figure 3: Exp-3. Estimated probability of predicting `Entail` for **random-premise** Levy/Holt conditioned on relative frequencies (Φ), with original ($I_{RandPrem}$) or generic ($I^{GenArg}_{RandPrem}$) entities. Predicting `Entail` is false positive hallucination (lower is better). Models hallucinate more often when test samples conform to the relative frequency bias ($\Phi_<$) than when not ($\Phi_>$).

sure with generic arguments, only 2 hypotheses are still predicted as `Attested` by GPT-3.5, whereas for LLaMA and PaLM, the numbers are also only 6.2% and 3.9%. Additionally, as with $I_{RandPrem}$, here the entailment label of each sample remains `No-Entail`, so any `Entail` prediction is false positive hallucination.

7.1 Results

We estimate the probabilities of models predicting `Entail` conditioned on the frequency label Φ, between $I_{RandPrem}$ and $I^{GenArg}_{RandPrem}$ settings, and present the results in Figure 3. We observe a clear and consistent rise of hallucination when samples conform to the bias. Namely, in case of $\Phi_<$, models are more likely to predict `Entail`, even though no semantic relation exists between P and H.

Between the two settings, with $I_{RandPrem}$, when entities are available, this effect is moderate. On the other hand, with $I^{GenArg}_{RandPrem}$ when entity-based memory is blocked, we observe a decrease in the overall level of hallucination, but the separation between $\Phi_<$ and $\Phi_>$ becomes more drastic, to 1.6x, 1.8x and 2.0x for LLaMA, GPT-3.5 and PaLM respectively. This indicates a tension between Λ and Φ: propositional memory may be used when available, and if not, the predicate pairing may be attended to more closely. Again, the Φ effect is observed across the three model families, revealing its root in the large-scale pre-training process, rather than model peculiarities or fine-tuning.

8 Impact of Bias on Performance

We have demonstrated two sources of hallucination by LLMs on inference tasks. We now assess their impact on model performance to quantify their risk.

We compare LLMs' performance between NLI subsets that are *consistent* or *adversarial* to each bias. A sample $P \vDash H$? is *consistent* with a bias when the prediction by the bias **agrees with** the gold entailment label; conversely, it is *adversarial* to a bias when the prediction by the bias **disagrees with** the label.

For example, "Google bought YouTube $\vDash$ Google owns YouTube" is *consistent* with the attestation bias of every model, because the conclusion *Google owns YouTube* is attested in every LLM's training data, and the sample label is `Entail`; "Apple owns Samsung $\nvDash$ Apple bought Samsung" is also *consistent*, because its conclusion is not attested and the sample label is `No-Entail`. The reverses of these two samples are *adversarial*, since their respective attestedness (unchanged) does not agree with the entailment labels (now flipped). For each subset, there is substantial representation in both Levy/Holt and RTE-1 (see appendix Table 9).

While earlier experiments inspected model textual responses to characterize behavior change, we now use area under the precision-recall curve (AUC) to summarize model performance over a tunable confidence threshold (scoring described in §4.2), which is better for measuring practical discriminative power. Following Li et al. (2022), we re-scale AUC values to normalize over the label distribution, yielding AUC_{norm} values that assign random classifiers 0% and perfect classifiers 100%.

We report results in Table 5. Under the standard inference task I, the performance drop from $\Lambda_{\text{CONSISTENT}}$ to $\Lambda_{\text{ADVERSARIAL}}$ is severe for all 3 LLMs: they deteriorate from very good classifiers to poor or even near-random ones.[8] This fragility from the *attestation bias* can be alleviated by masking entities with type-identifiers (condition I^{GenArg}), which reduces the performance drop.

On the other hand, with the generic arguments in I^{GenArg}, LLMs are forced to focus on the predicates in each proposition. As a result, the impact of the *relative frequency bias* is intensified. From the standard inference task I to I^{GenArg}, the average performance drop from the *cons.* to *adv.*

[8] We note Λ predictions could possibly be influenced by model-specific idiosyncrasies in prompt format. We provide an analysis in Appendix E, where we find no significant effect.

		Levy/Holt						RTE-1					
		Attestation (Λ)			Rel. Frequency (Φ)			Attestation (Λ)			Rel. Frequency (Φ)		
Model	**Task**	***cons.***	***adv.***	***diff.***	***cons.***	***adv.***	***diff.***	***cons.***	***adv.***	***diff.***	***cons.***	***adv.***	***diff.***
LLaMA	I	65.5	8.1	***-57.4***	42.1	32.3	***-9.8***	62.1	37.4	***-24.7***	55.5	51.7	***-3.8***
GPT-3.5	I	85.0	10.8	***-74.2***	53.5	43.2	***-10.3***	84.6	47.5	***-37.1***	77.6	43.4	***-34.2***
PaLM	I	79.1	31.5	***-47.6***	63.3	53.0	***-10.3***	87.1	83.4	***-3.7***	87.5	81.0	***-6.5***
LLaMA	I^{GenArg}	52.1	34.4	***-17.7***	55.3	34.9	***-20.4***	59.2	30.4	***-28.8***	51.7	39.4	***-12.3***
GPT-3.5	I^{GenArg}	67.1	18.8	***-48.3***	50.4	35.0	***-15.4***	80.1	56.4	***-23.7***	79.6	49.1	***-30.5***
PaLM	I^{GenArg}	58.1	46.6	***-11.5***	59.9	47.3	***-12.6***	78.1	84.4	*+6.3*	85.4	78.7	***-6.7***

Table 5: LLM performance on subsets where Λ/Φ is *consistent/adversarial* to entailment labels, measured with AUC_{norm} (0% = random chance performance). Decrease from $cons$ to adv subsets are shown in the *diff.* columns.

subsets w.r.t. Φ is widened from 10.1% to 16.1% for Levy/Holt and from 14.8% to 16.5% for RTE-1. The differences for Φ-consistency subsets are generally narrower than Λ-consistency subsets, possibly because the relative frequencies require generalizing from instances, and may be more difficult to capture, and potentially because frequency measures with Google N-gram are a crude estimate of the actual frequencies in LLM pre-train corpora.

9 Conclusion

Across several major LLM families and experimental settings, we demonstrate two important biases in the performance of LLMs on natural language inference tasks, which may also manifest in applied tasks as hallucination. Contrary to claims of LLM general reasoning capabilities, we show that much of this performance is achieved by (1) recall of relevant memorizations and (2) corpus-based biases like term frequency. Since these factors are reproduced in all models, we establish that they originate in LLM pre-training, and are not corrected during GPT-3.5 fine-tuning.

We conclude that LLMs, though powerful, use unsatisfactory tools for the basic tasks of language understanding and inference. We propose several approaches to control for these biases in evaluation, and ultimately conclude that further attention on alleviating these biases are needed, before LLMs may be trusted to reason robustly about language.

Limitations

In this paper, we have discussed two prominent sources of hallucination for LLMs in natural language inference tasks. We acknowledge that this is not an exhaustive search of all the sources, where further exploration should be done in future work.

We also note that after controlling for the factors discussed in this paper, there remains residual, unexplained performance on NLI tasks. This residual might be due to other undiscovered biases or possibly generalising inference capability. We leave further exploration of this residual to future work.

As discussed in Appendix A, we compared a range of popular LLM prompting techniques and selected the most promising approach. We acknowledge that there could potentially be other novel prompting techniques that could help the LLMs resist the influence of the biases discussed in this paper. We identify this as an open question and advocate for future research.

Ethical Considerations

This paper discusses two major sources of hallucination in LLM output when asked to perform natural language inference, which we note is a capability required of many downstream tasks such as summarization, question answering, etc. We show that users of LLMs may be subjected to faulty judgements if the content of their request overlaps with data in pretraining. However, it is difficult to ascertain for both a user or modeler exactly what is contained in pretraining data, or how this will interact with a user's query. Our proposed attestation query shows promise in detecting potential overlaps, but model responses in applications of these cases are not explored. Further, the relative frequency bias demonstrates a much more subtle problem of corpus distribution that is naturally inherent to model pretraining on human generated text.

In light of these, the potential harms of LLM use for drawing natural language inferences may include: offering inaccurate or irrelevant information to a user's query or contradiction of information provided in-context with a user's query.